

Group 15 elements [Pnicogens - VA]

⇒ Element	At. no	Ele. Conf'g
N	7	[He] $2s^2 2p^3$
P → most abundant element	15	[Ne] $3s^2 3p^3$
As	33	[Ar] $3d^{10} 4s^2 4p^3$
Sb	51	[Kr] $4d^{10} 5s^2 5p^3$
Bi	83	[Xe] $5d^{10} 6s^2 6p^3$
Mc	115	[Rn] $7s^2 7p^3$

→ ^{11th} Most abundant element in the Earth crust → Phosphorus (P).

→ G.E.C → $ns^2 np^3$.

→ Abundance order → $P > N > As > Sb > Bi$.

→ N_2 present in the air to the extent of 78% by volume and 80% by mass.

→ N_2 as nitrates KNO_3 , $NaNO_3$.

→ Except 'P' other elements are available in free state.

→ Phosphorus are present in animals, plants, living cells & bones.

* Phosphoproteins are present in milk and egg.

→ Phosphorus is available in & as

Phosphite : $(Ca_3(PO_3)_2)$

Fluorapatite : $3(Ca_3(PO_4)_2) \cdot (CaF_2)$

Chlorapatite : $3(Ca_3(PO_4)_2) \cdot (CaCl_2)$

Hydroxapatite : $3(Ca_3(PO_4)_2) \cdot (Ca(OH)_2)$

→ General Properties

Atomicity: → (AM)

N_2 - diatomic

P_4, As_4, Sb_4 - tetraatomic

Bi - monoatomic

→ Metallic nature (MN)

N and P → non metals

As and Sb → metalloids

Bi → metal

→ Electronegativity (EN)

$N > P > As > Sb \approx Bi$

→ Density ($d = M/V$):

$N(\text{at } 63K) < P < As < Sb < Bi$

→ Oxidation states:

The charge present on atom $-3, +3, +5$.

* The ability to exhibit +5 oxidation state, $Bi \rightarrow +3$

* $6s^2 6p^3$ due to inert pair effect.

* Bi only exhibits +5 in BiF_5 others As, Sb, Bi = -3 does not exhibit easily.

Nitrogen: [-3 to +5 oxidation state]

* $NH_3 = -3$

* $NH_2-NH_2 = -2$
(hydrazine)

* $NH_2OH = -1$
hydroxy amine

* $N_2 = 0$

→ Physical state:

N_2 - gas

P_4, As_4, Sb_4 - solid

Bi - solid

→ Covalent radius

$N < P < As < Sb < Bi$

→ I.P (Ionization potential)

$N > P > As > Sb > Bi$

→ Melting point (M.P):

$N < P < Bi < Sb < As$

→ Boiling point (B.P):

$N < P < As < Bi < Sb$

* NO_2 (or) $N_2O_4 = +4$
nitrogen dioxide

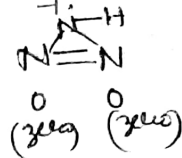
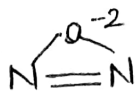
* N_2O_5 (or) $HNO_3 = +5$
nitric oxide (or) nitric acid

N_2O_3 (or) $HNO_2 = +3$

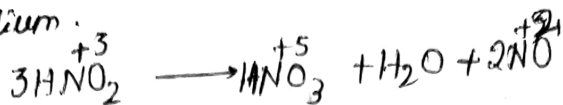
* $N_3H = -1/3$
Hydrazoic acid

* $N_2O = +1$
nitrous oxide

[+1 to +4 oxidation state]



→ Nitrogen(N) undergoes disproportionation in acidic medium.



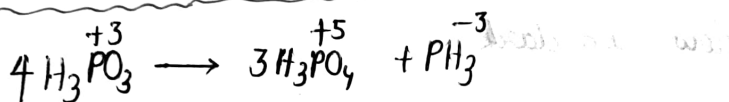
① $\text{H}_3\text{PO}_2 = +1$
hypophosphorous acid

② $\text{H}_3\text{PO}_3 = +3$
orthophosphorous acid

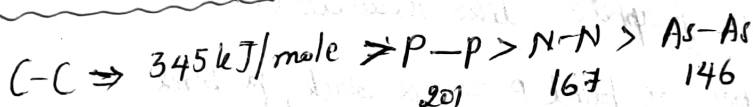
③ $\text{H}_3\text{PO}_4 = +5$
orthophosphoric acid

④ $\text{PH}_3 = -3$
phosphine

In acidic and alkaline medium:



Catenation capacity:



Allotropy:

→ Except "N" all other elements exhibit allotropy.

→ Phosphorus exhibit as

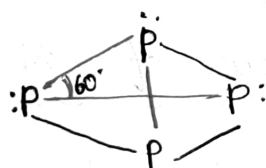
- i) white phosph.
- ii) Black P
- iii) Red P
- iv) violet P.

i) White P : → contains P_4 units

→ Translucent waxy solid

→ stored under water

→

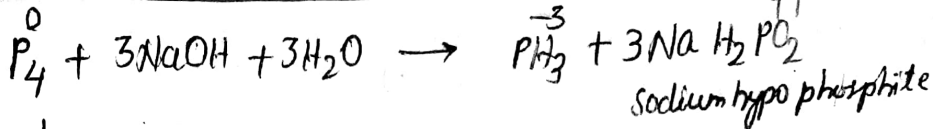


no. of P-P bonds = 6

→ This is most reactive form of phosphorus

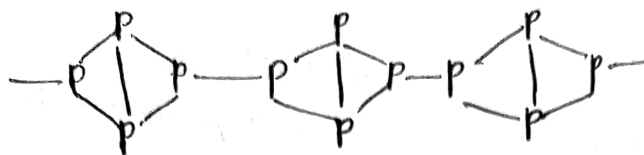
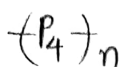
→ This is insoluble in water but readily soluble in CS_2 .

→ Glow in dark (chemiluminescence).



(disproportionation reaction) → undergoes both oxidation & reduction

ii) Red P :



* When white P (P₄) is heated at 573K then Red P (P₄) is formed
3 days

* Less reactive.

* Insoluble in H₂O & CS₂.

* Not glow in dark.

iii) Black P :

It has graphite like structure.

→ Two types of black P ⇒ α-black, β-black

* When red P $\xrightarrow{803K}$ α-black is formed.

* When white P $\xrightarrow{473K}$ β-black is formed.

* Thermodynamically most stable β-black P.

* Good conductor of electricity.

As : Metallic (or) green.

Non metallic (yellow)

Black

Sb : Metallic and explosive form

Anomalous behaviour of N :

Nitrogen (N) :

1. Has small size.

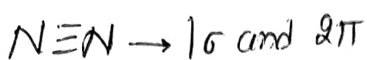
2. Has high I.P value

3. Has high EN value

4. Due to the absence of d-orbitals

nitrogen has ability to form π - π bonds with C and O, while other forms like phosphorus forms σ - π bonds.

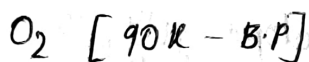
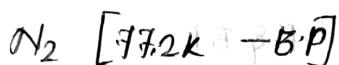
5) N_2 is diatomic.



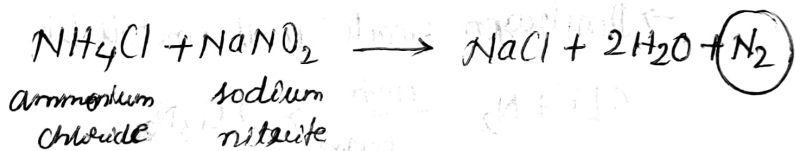
It has high bond energy (945.4 kJ/mole) due to it's high B.E, it is less reactive.

6) Preparations:

i) Commercially 'N' is prepared by air by liquefaction & fractional distillation of air.

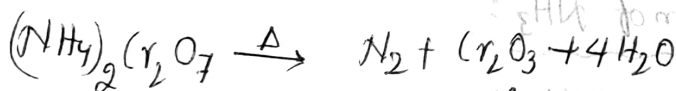


ii) In laboratory N_2 is prepared when ammonium chloride is heated with sodium nitrite.

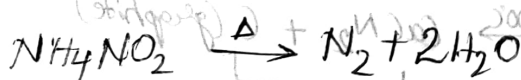


along N_2 , NO and HNO_3 to remove this it is passed through aq. soln of $H_2SO_4 + K_2Cr_2O_7$

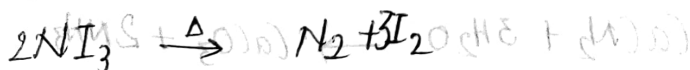
iii) Thermal decomposition:



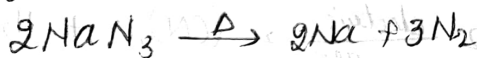
ammonium dichromate



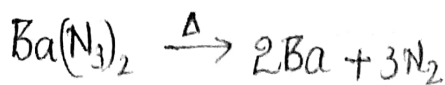
ammonium nitrite



iv) Pure form of N_2 is prepared from azide.



Sodium azide



Barium azide

v) From NH_3 :



vi) Properties: → Dinitrogen is colourless, odourless, non-toxic gas.

→ It has low solubility in water.

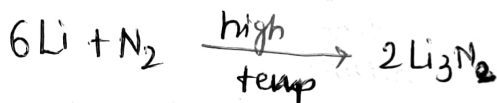
[23.2 cm³ per lit at 273K and 1 bar]

→ It has low F.P & B.P

→ It has two isotopes ${}^{14}_7\text{N}$ & ${}^{15}_7\text{N}$

→ N_2 has high B.E and less reactive
(945.4 kJ/mole)

→ Dinitrogen reacts with metals like

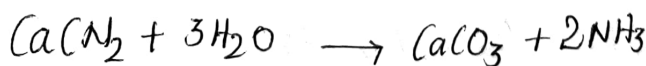
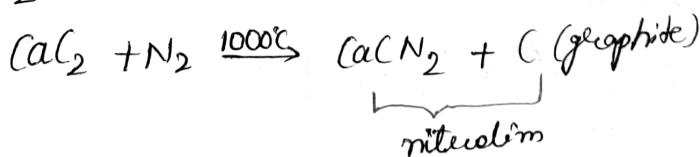
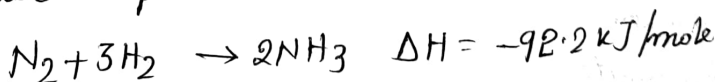


→ Dinitrogen reacts with non-metals like

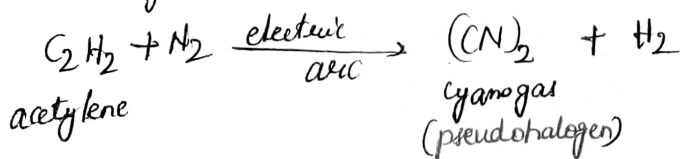


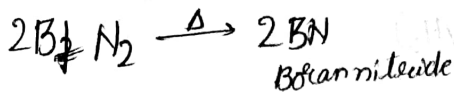
Preparation of NH_3 :

Haber's process:



Calcium cyanamide





Uses: $\rightarrow N_2$ used for preparation of NH_3 .

$\rightarrow$ It is used to produce inert atmosphere.

$\rightarrow$ liquid N_2 is used as refrigerant for biological system & food materials.

Hydrides: Group 15 elements forms MH_3 type hydrides

$\rightarrow NH_3$ ammonia

PH_3 phosphine

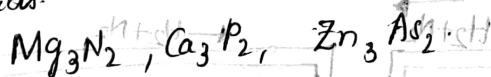
AsH_3 Arsenine

SbH_3 Stibine

BiH_3 Bismuthine

$\rightarrow$ Nitrogen also forms N_2H_4 (hydrazine) and N_3H (hydrazoic acid)

$\rightarrow$ These hydrides are prepared when metal nitrides, phosphates and arsenates are hydrolyzed with dil. acids.

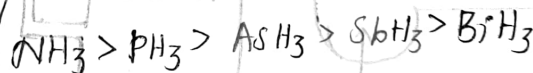


$\rightarrow$ Ease of formation

Thermal stability

Bond energy

Bond angle

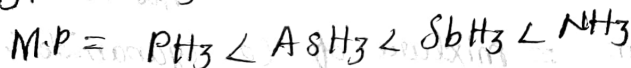
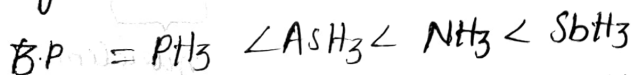


$\rightarrow$ These are Lewis bases.

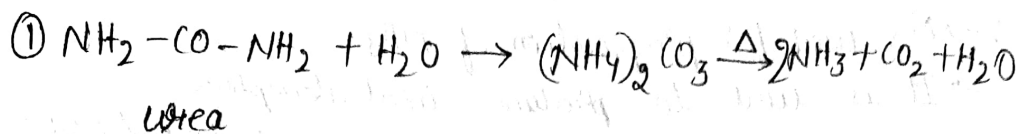
$\rightarrow$ Basic nature $= NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$.

$\rightarrow$ Reducing power $\left. \begin{array}{l} \text{Bond length} \end{array} \right\} = NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$

$\rightarrow NH_3$ has abnormally high BP due to intermolecular hydrogen bonds.



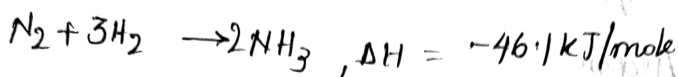
⇒ Preparation of ammonia (NH₃)



② Ammonium salts reacted with bases gives NH₃.



③ Haber's process



In 1:3 ratio the nitrogen & hydrogen are taken.

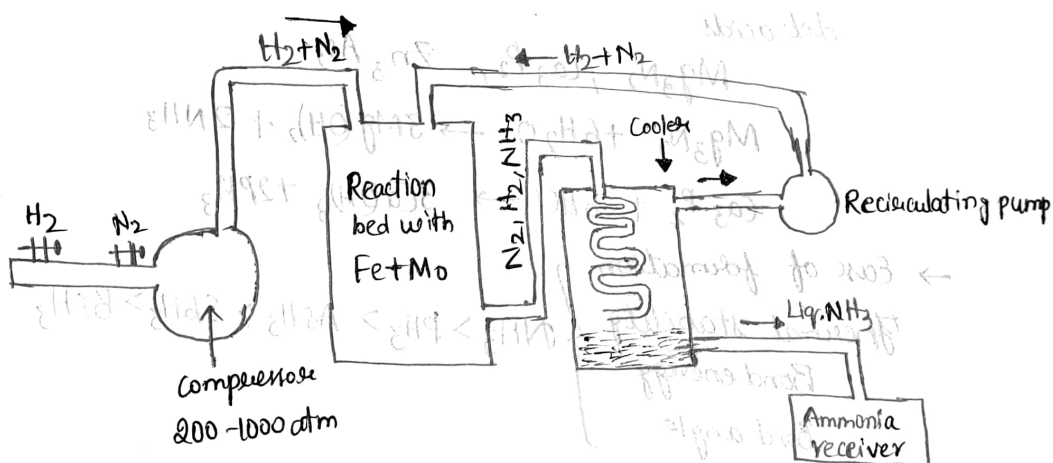
Acc. to Lechatelier the favourable conditions are

$$\text{temp} = 725 - 775 \text{ K}$$

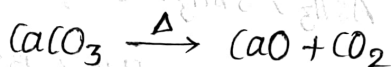
$$\text{press. (atm)} = 200 \text{ to } 500 \text{ atm}$$

$$\text{catalyst} = \text{Fe (or) mixture of } (\text{Fe}_2\text{O}_3 + \text{FeO}) + \text{Al}_2\text{O}_3 + \text{K}_2\text{O}$$

$$\text{promoter} = \text{Mo}$$

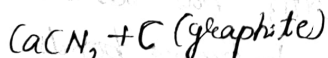
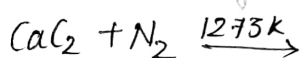


④ Cyanamide process:



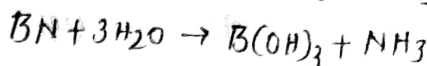
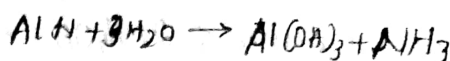
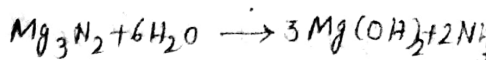
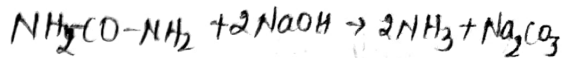
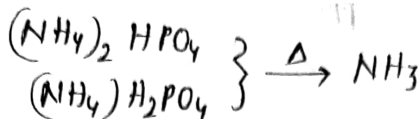
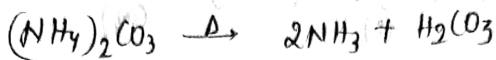
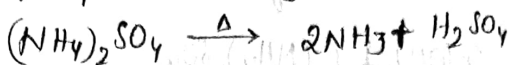
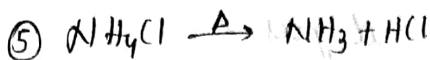
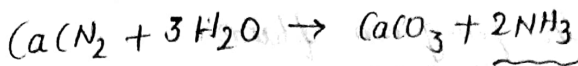
Calcium
carbide

CaCl_2 (or) CaF_2 are catalyst.



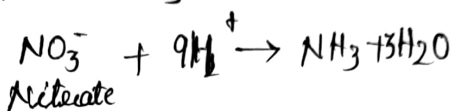
nitrilem

nitrilem = mixture of Ca cyanamide & graphite



boric acid

→ Reduction of nitrites and nitrates using Devarda's alloy:
[Ca/Al/Zn + NaOH]

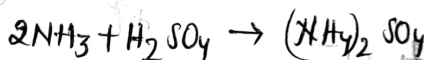
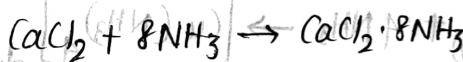


Nitrate



Nitrite

→ NH_3 can be dried by CaO (quicklime) anhyd. CaCl_2 & H_2SO_4 are dehydrating agent but can't be used due to it reacts with NH_3 .



→ Properties:

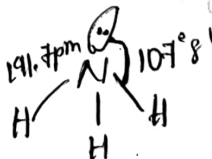
(i) Ammonium [is colourless and pungent odour gas.]

(ii) F.P = +195.2K B.P = 239.7K

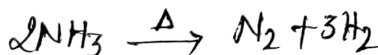
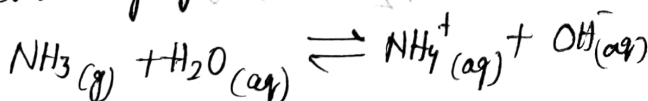
(iii) In liquid & solid state, NH_3 has strong hydrogen bonds

(iv)

shape - pyramidal

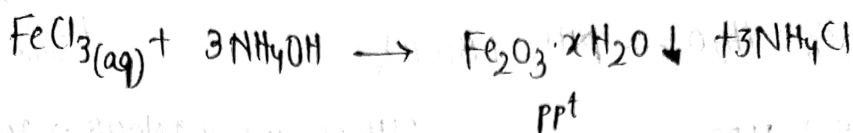
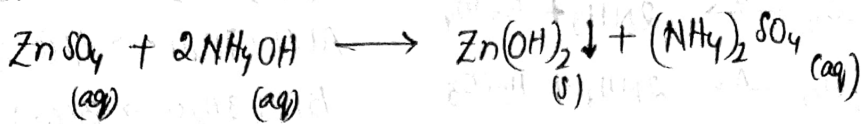


(v) It is highly soluble in water.

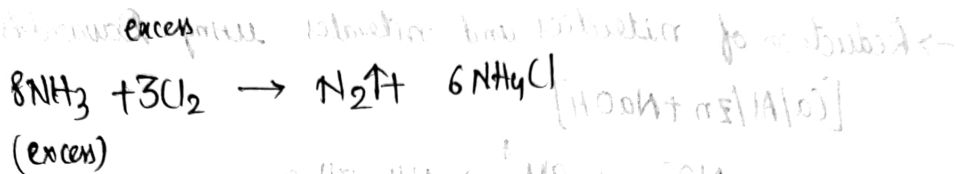
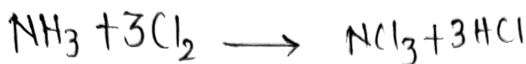


It burns with pale green flame.

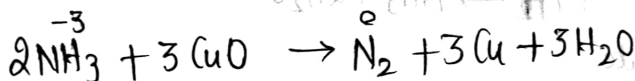
(vi) When it reacts with metal salts it gives



(vii) with Cl₂:



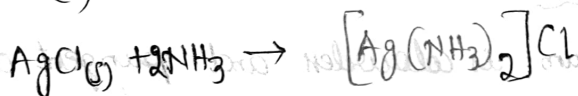
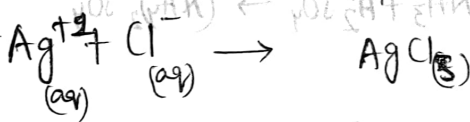
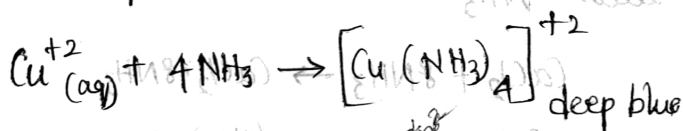
(viii) Oxidising:



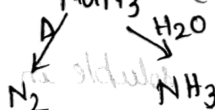
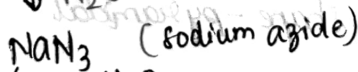
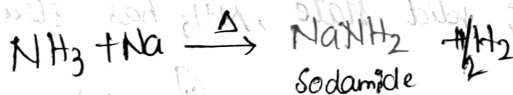
↓
Reducing agent:

PbO & CaCl₂

(ix) Forms complexes with transition metal ions:

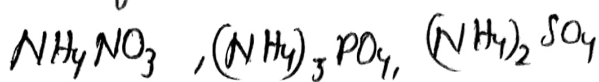


(x) with Na



Uses:

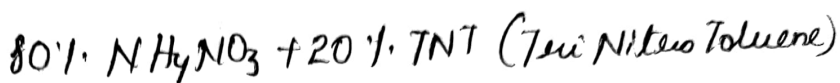
① Basic fertilizers



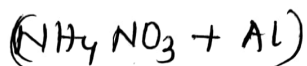
② Used to prepare HNO_3

③ Liquid NH_3 used as refrigerant.

④ Used to prepare explosives



Ammatol



Ammonal

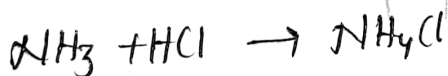
→ Test for NH_3 :

Brown ppt with Nessler's reagent



KI & HgCl_2 / NaOH (alkaline medium)

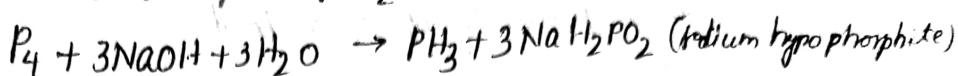
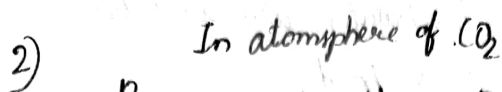
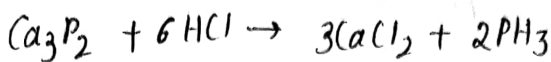
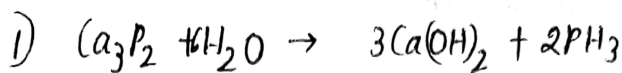
white fumes with HCl



Grey ppt with Hg_2Cl_2

11/4/23

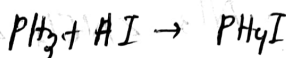
Phosphine (PH₃)



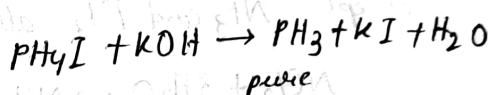
In laboratory phosphine is prepared by heating P_4 with NaOH in the inert atmosphere of CO_2 .

pure form of PH_3 is non-inflammable

impurities P_2H_4 and P_4 - highly inflammable



impure



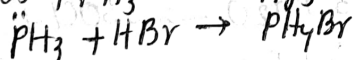
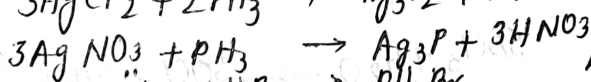
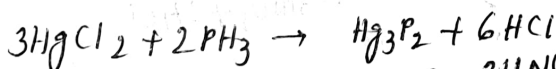
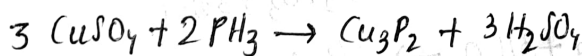
pure

PH_3 colourless gas

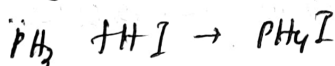
Rottenfish like smell

toxic gas

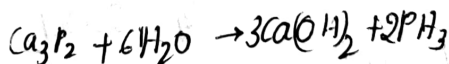
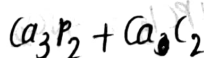
aqueous PH_3 on decomposition gives 'Red P' and H_2 .



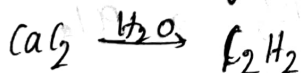
Lewis base - $\ddot{\text{P}}\text{H}_3$



Uses: $\rightarrow$ "Holme's signal"



Spontaneous combustion of phosphine used to produce Holme's signal. Mixture of $\text{Ca}_3\text{P}_2 + \text{CaCl}_2$ is used.



$\rightarrow$ Smoke screens are produced by phosphine.

Halides: Group 15 elements form two types of halides.

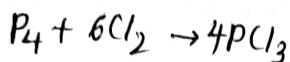
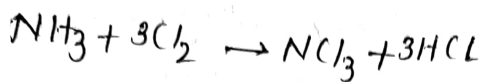
(MX₃) Trihalides and pentahalides (MX₅)

Trihalides NCl₃, PCl₃, PF₃, NF₃

Except NF₃ remaining 'N' trihalides are unstable.

1) Low polarity of N-X bond

2) "N-X bond is weak"

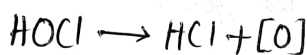
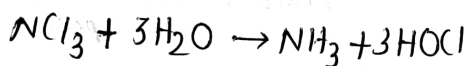


Covalent nature except BiF₃

Lewis bases $\text{NX}_3 > \text{PX}_3 > \text{AsX}_3 > \text{SbX}_3 > \text{BiX}_3$

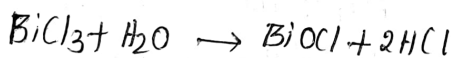
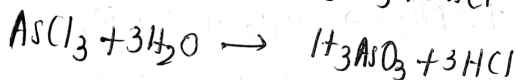
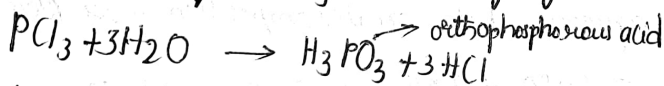
Basic nature $\text{NF}_3 < \text{NCl}_3 < \text{NBr}_3 < \text{NI}_3$

Except NF₃ and PF₃ all halides undergo hydrolysis.



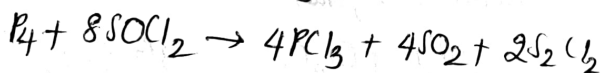
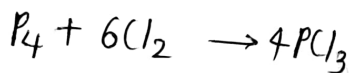
Coloured + [O] → colourless nascent oxygen [O]

Ag. NCl₃ - Bleaching agent

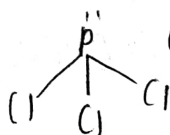


Oxoanions are formed (SbO⁺Cl⁻)

PCl₅:



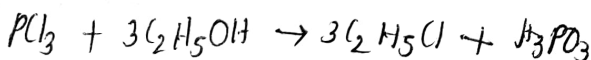
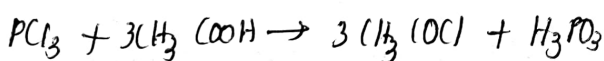
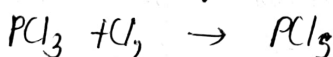
P - sp³



(pyramidal)

Thionyl chloride

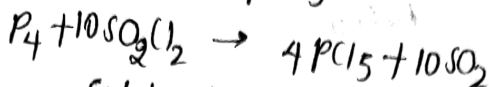
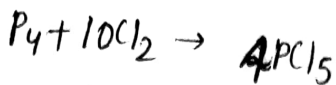
white oily liquid



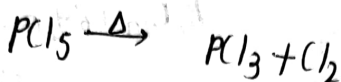
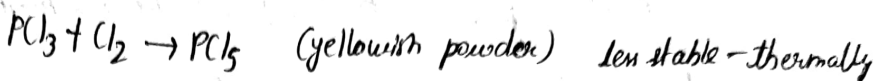
Pentahalides - MX_5

$PCl_5, PF_5, SbCl_5, BiF_5$ - pentahalide

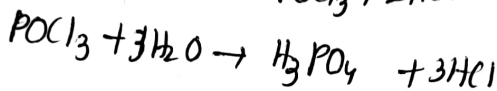
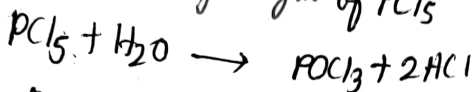
* 'N' doesn't form NX_5 due to absence of vacant d-orbitals.



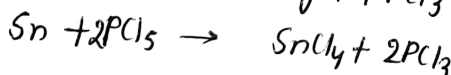
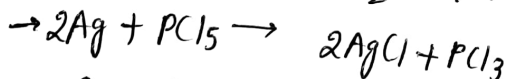
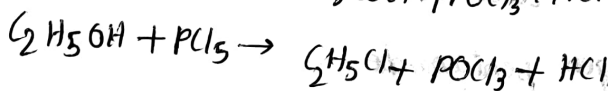
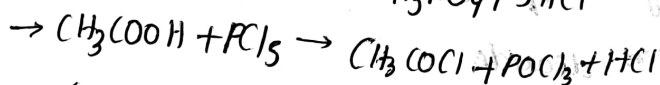
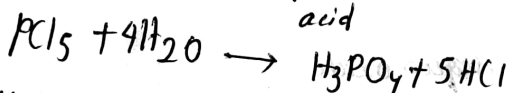
Sulphonyl
chloride



Hydrolysis of PCl_5



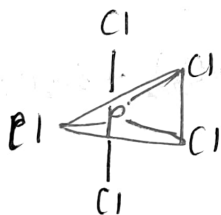
phosphoric
acid



PCl_5

p - sp^3d

Trigonal bipyramidal



In solid state PCl_5 exists as $[PCl_4]^+ [PCl_6]^-$
 $\downarrow \quad \downarrow$
 $sp^3 \quad sp^3d^2$
 Tetrahedral octahedral

PBr_5 exists as $[PBr_4]^+ [Br]^-$

NOTE: PI_5 doesn't exist due to large size of iodine and small electronegative difference b/w phosphorus and iodine.

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Oxides:

Trioxides

Pentaoxides



N_2O_5 strong oxidizing agent



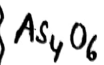
Acidic



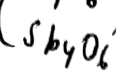
Acidic



Dimeric
exists as

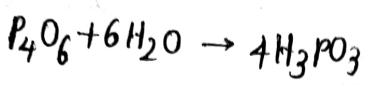


Amphoteric

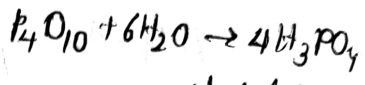


Basic

Bi_2O_5 pentaoxides are most acidic



phosphorous
acid



phosphoric acid

Oxides of nitrogen

N_2O - nitrous oxide

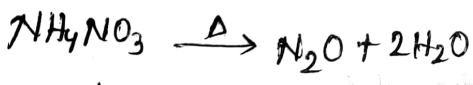
NO - nitric oxide

N_2O_3 - nitrogen sesquioxide

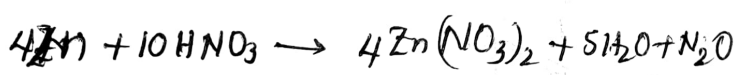
NO_2 - nitrogen dioxide

N_2O_5 - nitrogen pentoxide

Nitrous oxide (N_2O)



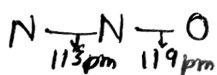
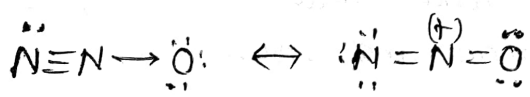
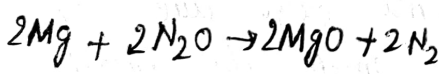
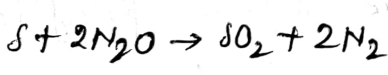
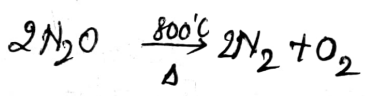
ammonium
nitrate



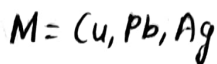
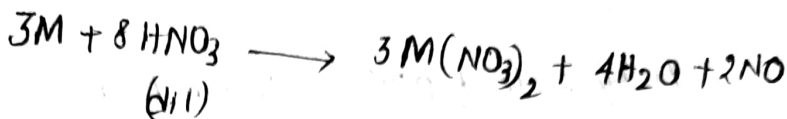
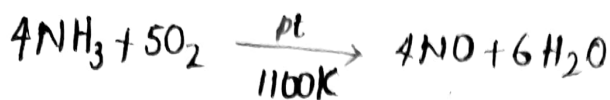
dil

→ "laughing gas".

→ colourless and neutral oxide



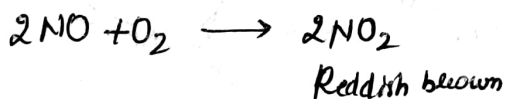
Nitric oxide (NO):



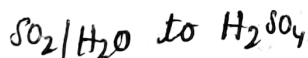
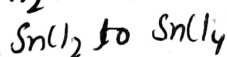
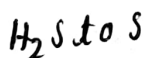
→ Colourless gas

→ neutral oxide

→ odd e molecule - paramagnetic nature



It oxidises

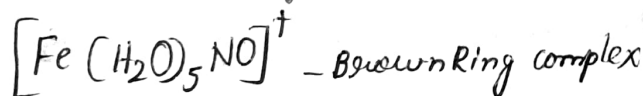


	NO	NO^+ (Nitrosonium ion)
Bond order	2.5	3
Bond length	1.15 Å	1.06 Å

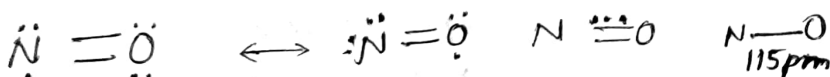
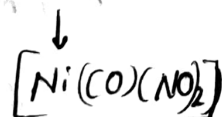
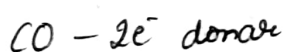
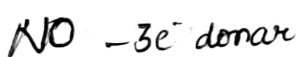
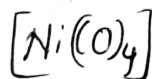
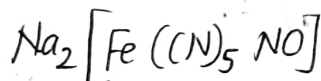
Bond order $\propto \left(\frac{1}{\text{Bond length}} \right)$



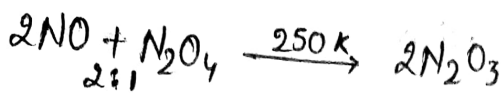
nitrosyl chloride



Fe +1
Oxidation state



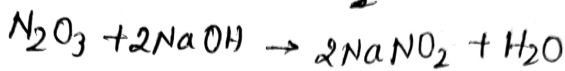
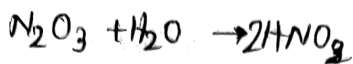
Nitrogen sesquioxide (N_2O_3)



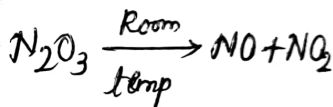
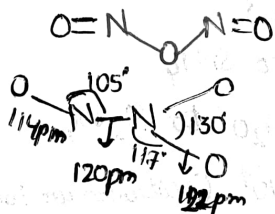
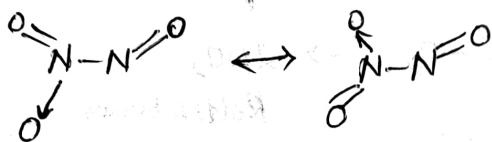
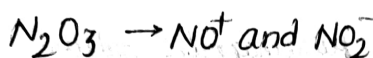
→ Blue solid (or) liquid

→ Acidic oxide

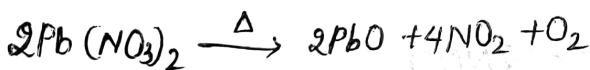
Anhydride of HNO_2 (Nitrous Acid)



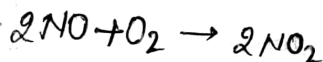
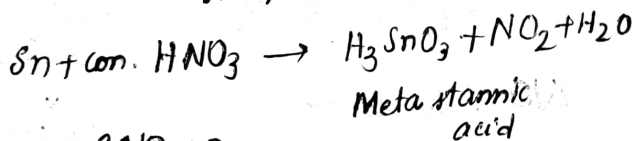
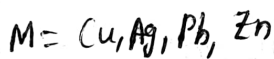
Self ionisation



Nitrogen dioxide (NO_2)

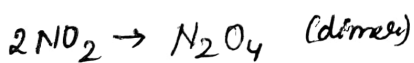


Lead nitrate



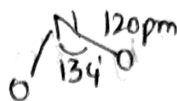
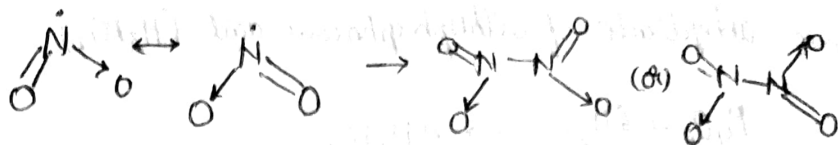
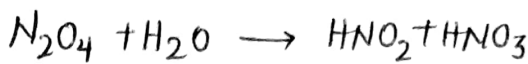
→ Reddish brown

→ odd e^- molecule, paramagnetic.



→ colourless, diamagnetic

NO_2 is mixed anhydride of HNO_2 and HNO_3 .

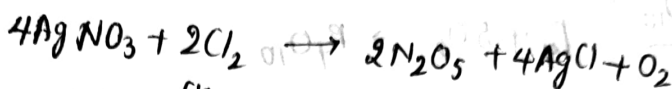


In liquid state it exists as N_2O_4 $\text{NO}^+ \text{NO}_3^-$

Nitrogen pentoxide (N_2O_5):



P_4O_{10} - dehydrating agent.

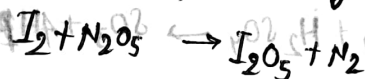


N_2O_5 - strong acidic oxide

Acidic nature $\Rightarrow \text{N}_2\text{O} < \text{NO} < \text{N}_2\text{O}_3 < \text{NO}_2 < \text{N}_2\text{O}_5$
 $\begin{matrix} +1 & +2 & +3 & +4 & +5 \end{matrix}$

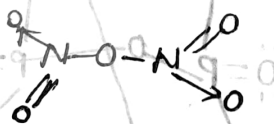
Anhydride of HNO_3 ($\text{N}_2\text{O}_5 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3$)
 nitric acid

$\rightarrow \text{N}_2\text{O}_5$ - strong oxidising agent



$\rightarrow$ In vapour state - covalent, in solid state - ionic

N_2O_5 as NO_2^+ and NO_3^-



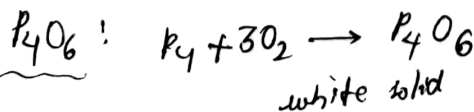
13/4/23

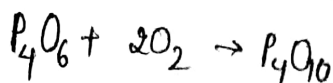
Oxides of phosphorous

Two types

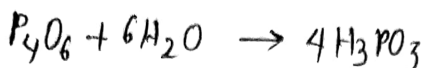
P_4O_6 (phosphorous trioxide)

P_4O_{10} (phosphorous pentoxide)

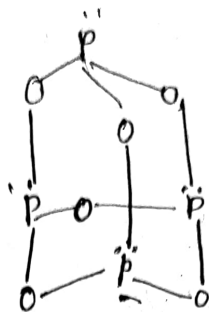




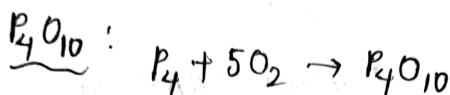
P_4O_6 anhydride of orthophosphorous acid (H_3PO_3)



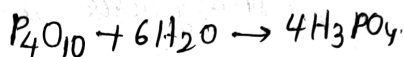
In P_4O_6 each "P" by 3 "O" atoms



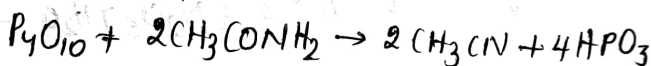
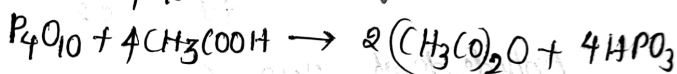
Total no of P-O-P bonds is 6
no of covalent bonds - 12



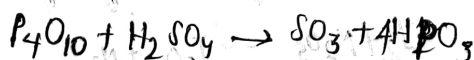
Anhydride of phosphoric acid (H_3PO_4)



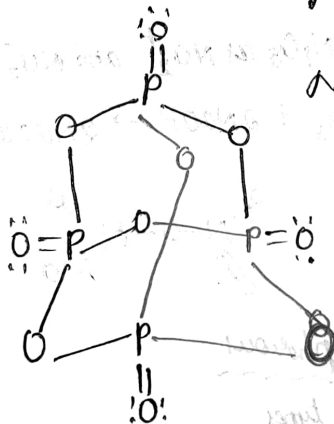
P_4O_{10} - Good dehydrating agent



acetamide



Each 'P' surrounded by 4 'O' atoms



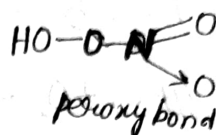
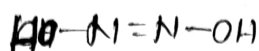
No. of P-O-P bonds - 6
Total no. of covalent bonds = 16

es/A/31

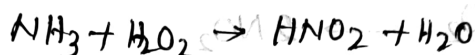
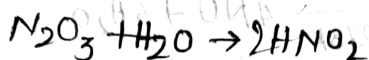
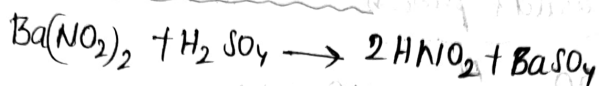
OXO acids of nitrogen

- 1) Hyponitrous acid - $H_2N_2O_2$ (or) HNO
- 2) Nitrous acid - HNO_2
- * 3) Nitric acid - HNO_3
- 4) Perinitic acid - HNO_4

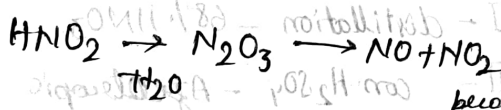
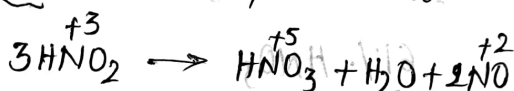
Hypo nitrous acid $H_2N_2O_2$ Perinitic acid (HNO_4)



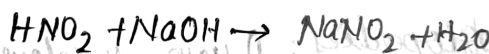
Nitrous acid (HNO_2)



Properties : Unstable, weak acid



HNO_2 salts - nitrites



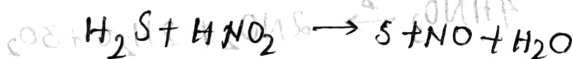
sodium nitrite

HNO_2 - Good oxidizing agent

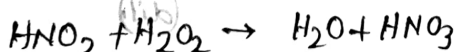
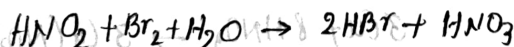
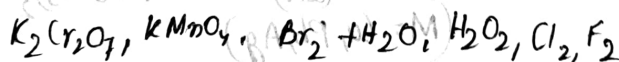
Reduced to N_2O (or) NO and ~~reduced~~ H_2S to S

I^- to I_2

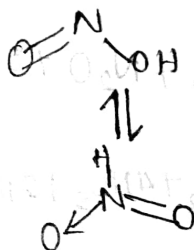
Fe^{+2} to Fe^{+3}



HNO_2 - Reducing agent oxidised to NO_3^-

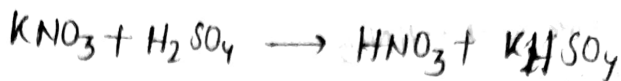


Tautomeric structure

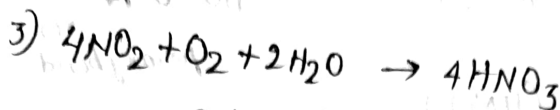
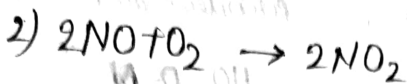
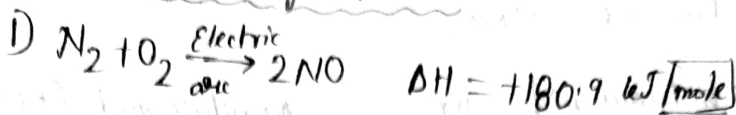


Nitric acid (HNO_3)

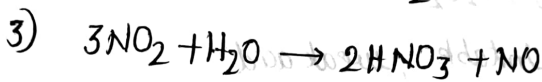
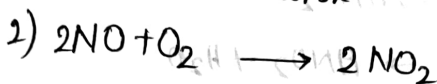
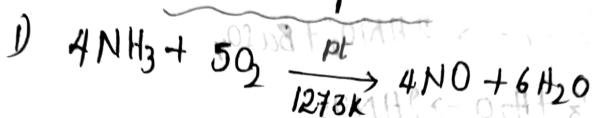
In Lab NaNO_3 , KNO_3 , NH_4NO_3



Beckland - Ede process



Ostwald's process



61% HNO_3

Step I - distillation - 68% HNO_3

Step II - con. H_2SO_4 - Azeotropic - 98% HNO_3

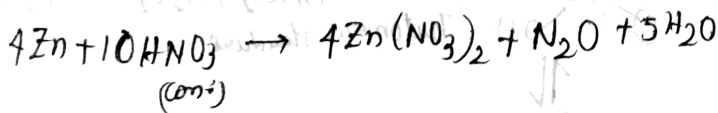
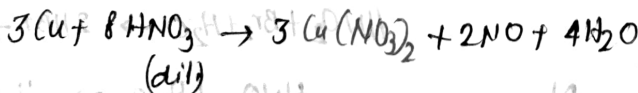
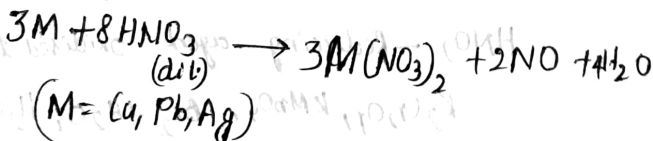
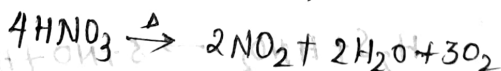
Step III - HNO_3 in freezing mixture - 100% pure

Properties: 1) HNO_3 is colourless liquid.

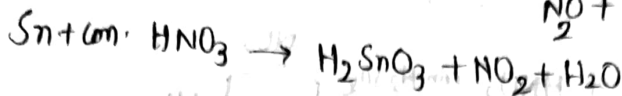
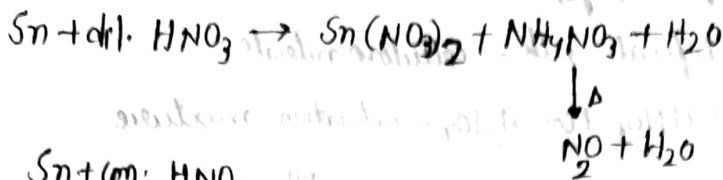
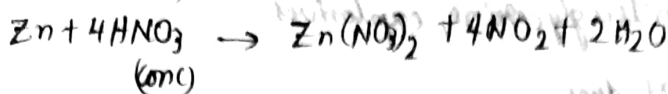
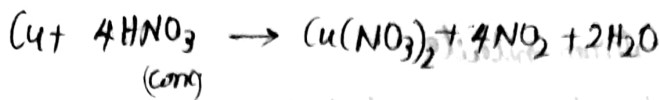
2) F.P - 231.4 K

3) B.P - 355.6 K

4) Good oxidising agent



M = Cu, Pb, Ag, Hg, Zn

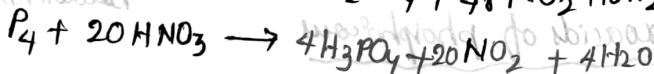
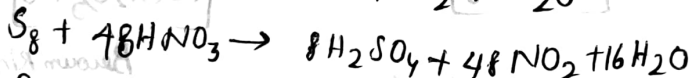
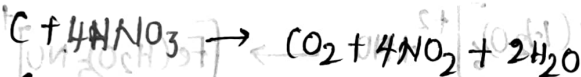
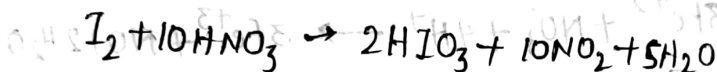


Metastannic acid

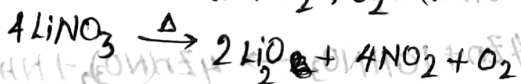
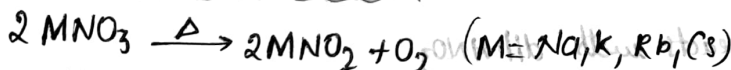
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(Cr, Ni, Al, Fe) metals do not react with HNO_3 due to passivity (formation of oxide layer).

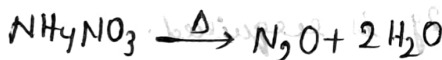
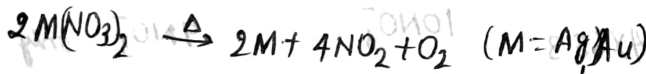
with non metals



Heating effect of nitrates

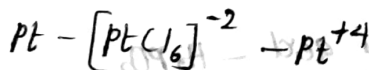


(M = Ca, Mg, Sr, Ba, Ra)



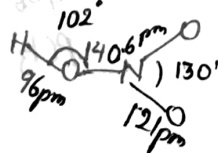
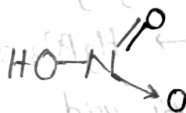
Aqua regia

1 part HNO_3 + 3 part of HCl



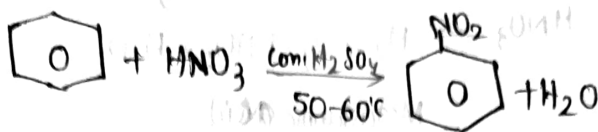
NO_3^- reduces to NO

structure



Uses: Basic calcium nitrate

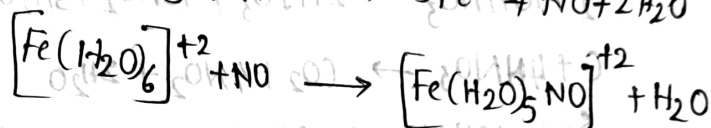
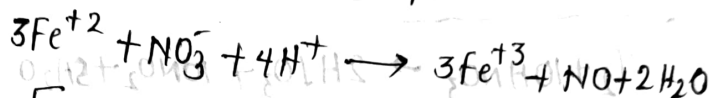
- TNT, nitroglycerine, explosives
- perfumes, dyes
- artificial silk - cellulose nitrate
- $\text{con. HNO}_3 + \text{con. H}_2\text{SO}_4$ - nitration mixture



NO_3^- ion test

Brown Ring test

Salt soln + $\text{FeSO}_4 + \text{con. H}_2\text{SO}_4$

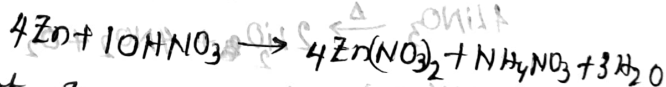


Brown Ring complex

Oxoacids of phosphorous

Two series of oxoacids

Q) When Zn reacts with dil. HNO_3



amount of Zn required to required to make
of HNO_3 (at wt of Zn) 65.3



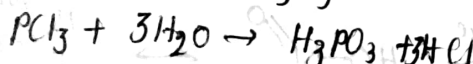
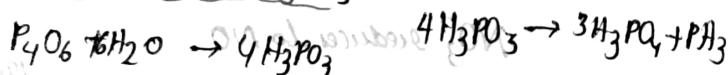
= 261.2g is required.

1) Phosphorous acid series

2) phosphoric acid series

1) Phosphorous acid series

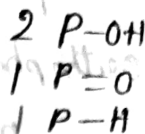
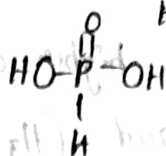
Orthophosphorous acid - H_3PO_3



Dibasic acid

two types of salts $H_2PO_3^-$ - primary phosphite

HPO_3^{2-} secondary phosphite



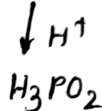
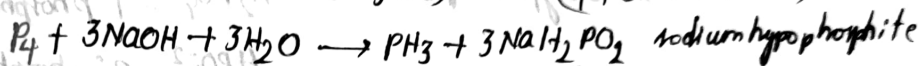
OH^- group is responsible for basicity of acids & due to P-H bonds it acts as good reducing agent.



2) Metaphosphorous acid (HPO_2)

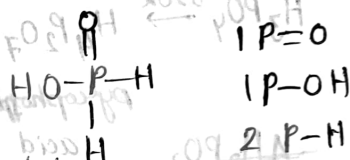


3) Hypophosphorous acid (H_3PO_2) (+1 oxidation of P)



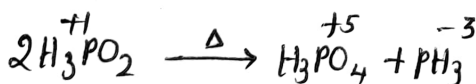
Monobasic acid

Basicity = 1

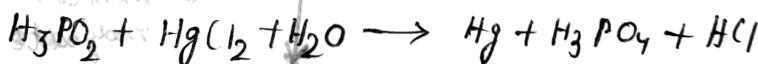
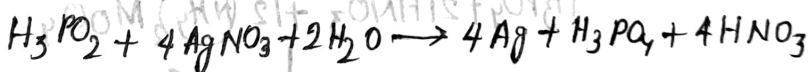


salts - hypophosphite

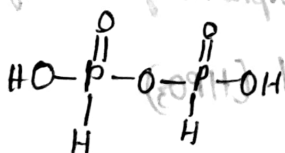
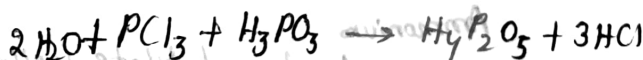
Action of heat



Good reducing agent



Pyrophosphorous acid ($H_4P_2O_5$) (P.O. state +3)

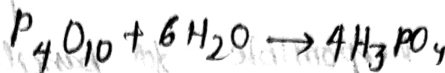


two P=O
two P-OH
two P-H
one P-O-P

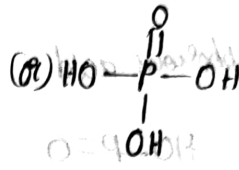
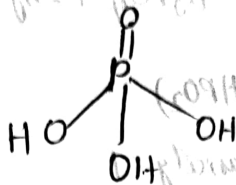
phosphoric acid series

"P" exhibit higher O. state

1) Orthophosphoric acid (H_3PO_4) (+5 O.S)



→ Tetrabasic acid Basicity = 3



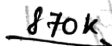
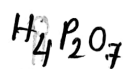
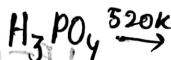
3 P-OH
1 P=O

→ 3 types of salts

$H_2PO_4^-$ - 1st phosphate (acidic) NaH_2PO_4

HPO_4^{2-} - 2nd phosphate (acidic) Na_2HPO_4

PO_4^{3-} - 3rd phosphate Na_3PO_4

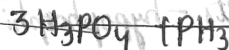


pyrophosphoric acid

metaphosphoric acid

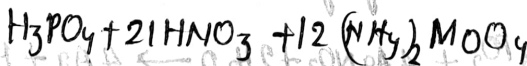


acid



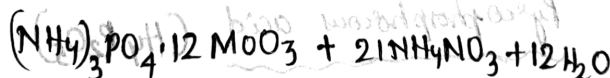
Quantitative analysis

O.C → H_3PO_4
Organic compound



Ammonium

molybdate

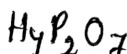
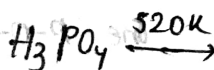


Ammonium

phosphomolybdate (yellow)

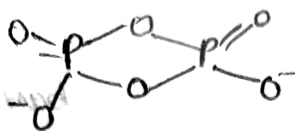
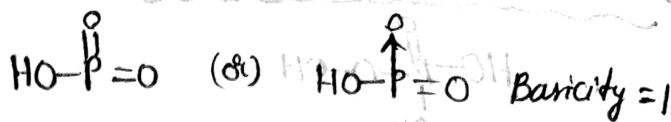
Metaphosphoric acid (HPO_3)

Glacial

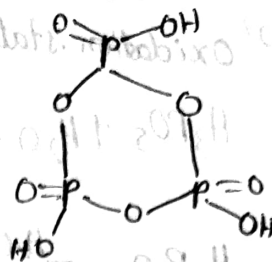


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phosphorous acid + Br_2 heated in sealed tube HPO_3 is formed.



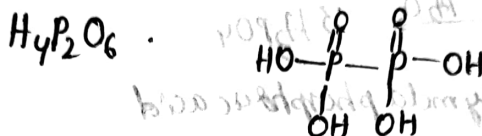
Dimer



$(\text{HPO}_3)_3$

Trimer

Hypophosphoric acid ($\text{H}_4\text{P}_2\text{O}_6$)

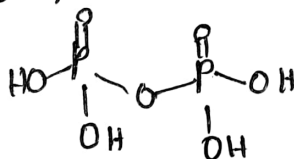
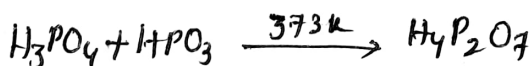


Basicity = 4

'P' Oxidation state = +4

no. of P-OH = 4

pyrophosphoric acid ($\text{H}_4\text{P}_2\text{O}_7$)



Basicity = 4

o. state = +5

P-OH = 4

P=O = 2

P-O-P = 1

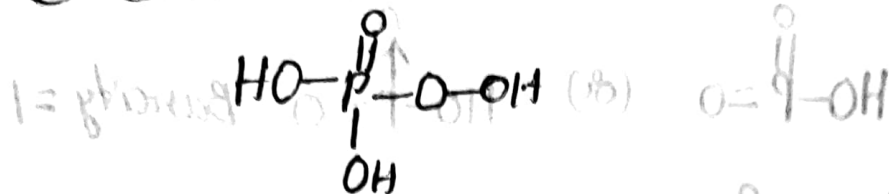
$\rightarrow$ In these P is bonded at least one OH group

$\rightarrow$ OH is responsible for basicity of acid

$\rightarrow$ 'ox' acids consists of P-H bonds responsible for reducing property.

$\rightarrow$ 'ic' acids does not contain P-H bonds.

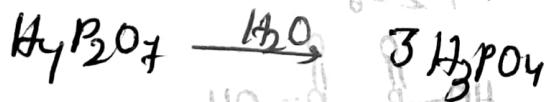
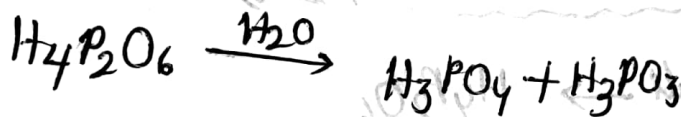
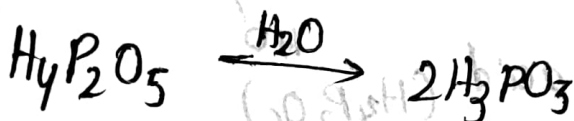
(H₃PO₅) Peroxophosphoric acid



'P' oxidation state +5



NOTE:



poly meta phosphoric acid

